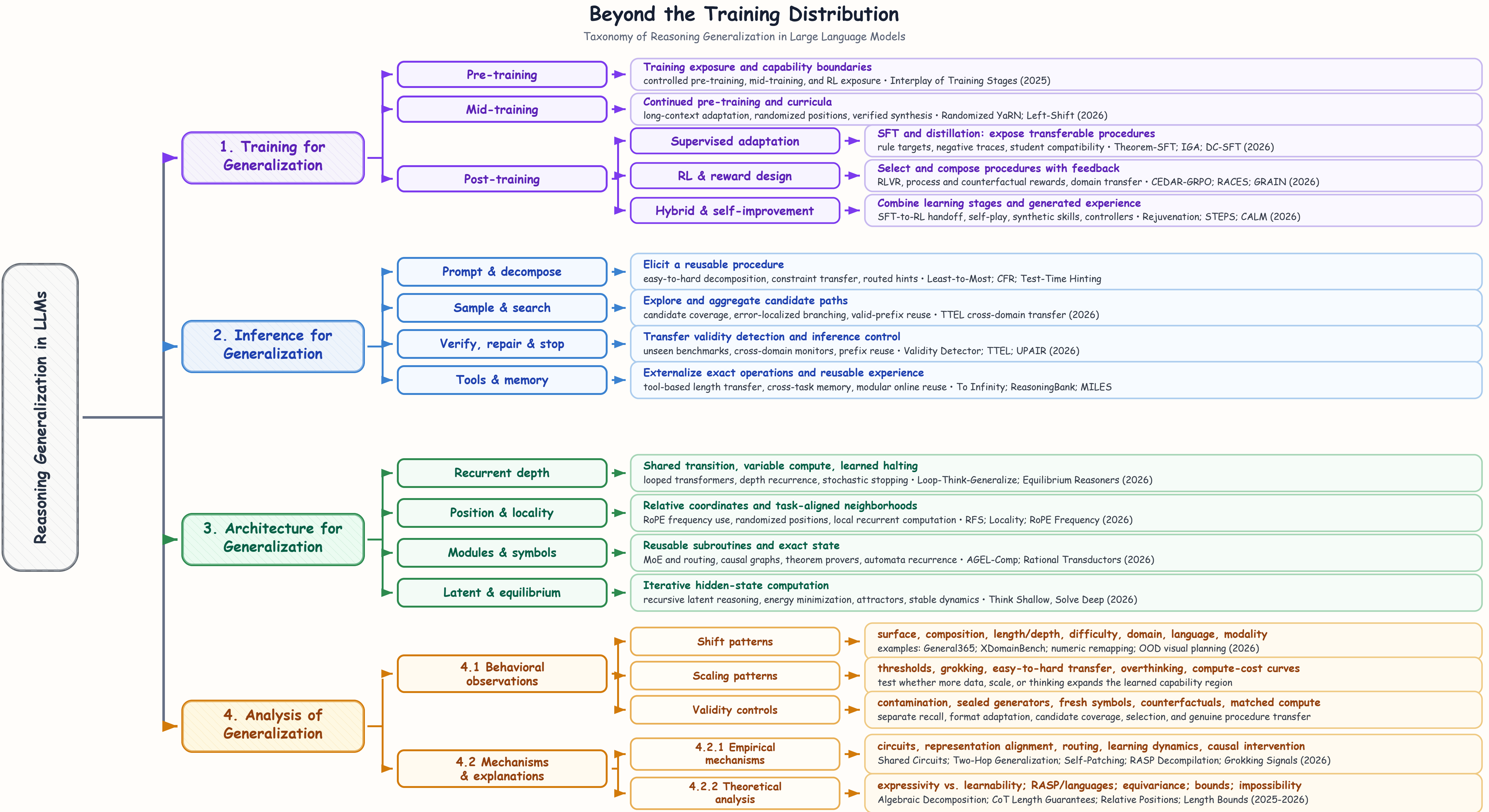




Beyond the Training Distribution

Taxonomy of Reasoning Generalization in Large Language Models

1. Training for Generalization

Pre-training

Mid-training

Post-training

Training exposure and capability boundaries

controlled pre-training, mid-training, and RL exposure • Interplay of Training Stages (2025)

Continued pre-training and curricula

long-context adaptation, randomized positions, verified synthesis • Randomized YaRN; Left-Shift (2026)

Supervised adaptation

SFT and distillation: expose transferable procedures

rule targets, negative traces, student compatibility • Theorem-SFT; IGA; DC-SFT (2026)

RL & reward design

Select and compose procedures with feedback

RLVR, process and counterfactual rewards, domain transfer • CEDAR-GRPO; RACES; GRAIN (2026)

Hybrid & self-improvement

Combine learning stages and generated experience

SFT-to-RL handoff, self-play, synthetic skills, controllers • Rejuvenation; STEPS; CALM (2026)

2. Inference for Generalization

Prompt & decompose

Sample & search

Verify, repair & stop

Tools & memory

Elicit a reusable procedure

easy-to-hard decomposition, constraint transfer, routed hints • Least-to-Most; CFR; Test-Time Hinting

Explore and aggregate candidate paths

candidate coverage, error-localized branching, valid-prefix reuse • TTEL cross-domain transfer (2026)

Transfer validity detection and inference control

unseen benchmarks, cross-domain monitors, prefix reuse • Validity Detector; TTEL; UPAIR (2026)

Externalize exact operations and reusable experience

tool-based length transfer, cross-task memory, modular online reuse • To Infinity; ReasoningBank; MILES

3. Architecture for Generalization

Recurrent depth

Position & locality

Modules & symbols

Latent & equilibrium

Shared transition, variable compute, learned halting

looped transformers, depth recurrence, stochastic stopping • Loop-Think-Generalize; Equilibrium Reasoners (2026)

Relative coordinates and task-aligned neighborhoods

RoPE frequency use, randomized positions, local recurrent computation • RFS; Locality; RoPE Frequency (2026)

Reusable subroutines and exact state

MoE and routing, causal graphs, theorem provers, automata recurrence • AGEL-Comp; Rational Transductors (2026)

Iterative hidden-state computation

recursive latent reasoning, energy minimization, attractors, stable dynamics • Think Shallow, Solve Deep (2026)

4. Analysis of Generalization

4.1 Behavioral observations

4.2 Mechanisms & explanations

Shift patterns

Scaling patterns

Validity controls

4.2.1 Empirical mechanisms

4.2.2 Theoretical analysis

surface, composition, length/depth, difficulty, domain, language, modality

examples: General365; XDomainBench; numeric remapping; OOD visual planning (2026)

thresholds, grokking, easy-to-hard transfer, overthinking, compute-cost curves

test whether more data, scale, or thinking expands the learned capability region

contamination, sealed generators, fresh symbols, counterfactuals, matched compute

separate recall, format adaptation, candidate coverage, selection, and genuine procedure transfer

circuits, representation alignment, routing, learning dynamics, causal intervention

Shared Circuits; Two-Hop Generalization; Self-Patching; RASP Decompilation; Grokking Signals (2026)

expressivity vs. learnability; RASP/languages; equivariance; bounds; impossibility

Algebraic Decomposition; CoT Length Guarantees; Relative Positions; Length Bounds (2025-2026)